PROJ 14: A lift

Design a lift controller, which allows for moving of a lift platform between 7 levels (0..6). The lift should accelerate after leaving a start platform and decelerate before reaching the destination platform. The level is chosen from a key pad. The display should show the actual position. There is also a coarse acoustic distance meter on board.

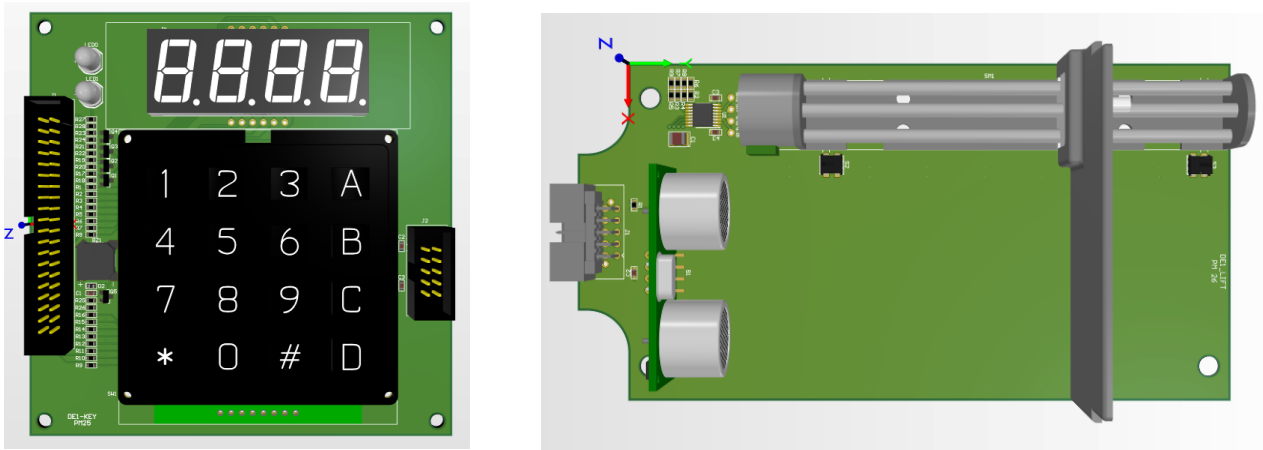

Figure 1. The key pad and the lift mechanism.

Details

1. The lift is driven by a step motor via a DRV8428PWPR driver. The FPGA sends 4 signals to the driver: STEP, DIR, EN and nSLEEP. Make sure the EN and nSLEEP are ON only while moving the motor, otherwise the motor gets hot.
2. The full speed of the motor is 2000 steps/s
3. The full distance from level 0 to level 6 is ~6550 steps.
4. The acceleration and deceleration to reach the full speed should take 2-3s each. The destination level should be reached at the lowest speed.
5. The position of the platform should be based on the motor step counter. I propose level 0 to be 0 steps while level 6 -6000 steps. The display on the key pad should show the accurate position in steps. A simple choice of the destination platform could be done by pressing buttons 0..6. A more elaborated method is to choose the desired fine level from 0 to 6550.
6. There are two end-point switches, connected electrically to one signal.
7. Before using the lift, the position needs to be calibrated. The platform should gently move toward the 0-level end-point switch and stop immediately after the switch is pressed. While reaching the end point the lift should move at the lowest speed.
8. On the lift board there is an acoustic distance meter HC-SR04. It may be helpful to establish a course position of the platform in cases where the position given by the step counter is not correct. The distance between the sensor and the platform (expressed in mm) can be shown on the DE1 evaluation board's LED display.
9. Reaching the destination level can be signaled by a beeper. The resonant frequency is 2.7 kHz. The idle level of the beeper signal should be 0.
10. Forcing the motor beyond the terminal position does not lead to a motor failure, however it's better to avoid it. Include an emergency STOP button in the design.
11. The key pad is a simple row-column matrix without no-bouncing guarantee. 4 row and 4 column signals are weakly pulled-up to 3.3V. To find out which key is pressed, the columns are driven to 0 in turn, while sensing the row signals.
12. The display is of a multiplexed common cathode type. Digits are driven to 1 in turn, while

sending the corresponding 7-segments code for the driven digit. The segment lights when driven to 1.

13. One can use 1 kHz clock for driving the key-pad column sequencer and display digit sequencer
14. For conversion between BCD and Binary use a double dabble method. Do not use dividers. https://en.wikipedia.org/wiki/Double_dabble